Investment Management — Practice Problems with Full Solutions

One worked problem of every type you could be asked · Prof. Anil Misra mid-term

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How to use this. Cover the solution, attempt the problem with pen and paper, then check. The numbers are fresh (not from your paper) but the *methods* are identical. In the real exam, write the same way these solutions are written: **interpret, state the approach, show the formula and working, then conclude** — that is where 90% of the marks are.

A. Risk, Return and Choosing a Stock (Q1 type)

Problem 1 — Full stock-comparison

You are given six monthly closing prices for two stocks and the market index:

Month	Alpha Ltd	Beta Ltd	Index
0	250	480	15000
1	260	500	15300
2	255	490	15200
3	275	520	15600
4	270	510	15450
5	290	505	15800
6	300	540	16000

(a) Which stock is the better investment? (b) What is the worst-case monthly loss on Rs 1,00,000 in each, at 95% confidence?

Solution.

Interpret: we must compare on a **risk-adjusted** basis, not on return alone, and express worst-case loss as VaR.

Step 1 — returns, using $R_t = (P_t - P_{t-1})/P_{t-1}$:

- Alpha: +4.00%, −1.92%, +7.84%, −1.82%, +7.41%, +3.45%
- Beta: +4.17%, −2.00%, +6.12%, −1.92%, −0.98%, +6.93%
- Index: +2.00%, −0.65%, +2.63%, −0.96%, +2.27%, +1.27%

Step 2 — statistics (mean; sample SD with $n-1$; R-T-R = mean/SD; β = SLOPE = Cov/Var; VaR $= -$mean $+ 1.645$ SD):

Measure (monthly)	Alpha	Beta	Index
Average return	3.16%	2.05%	1.09%
Std deviation (risk)	4.28%	4.15%	1.54%
R-T-R = Mean/SD	**0.739**	0.494	0.708
Beta vs Index	2.71	1.75	1.00
Correlation with Index	0.975	0.647	1.00
VaR 95% = −Mean+1.645·SD	3.87%	4.78%	1.44%
VaR on Rs 1,00,000	Rs 3,873	Rs 4,780	Rs 1,443

(a) Alpha wins on every count: higher return (3.16% vs 2.05%), higher **R-T-R** (0.739 vs 0.494 — the key criterion), *and* the lower worst-case loss. **Choose Alpha Ltd.** (Note its beta of 2.71 means it is aggressive — it amplifies market moves ~2.7×; that is where its extra return comes from.)

(b) Worst-case monthly loss at 95%: **Alpha ≈ Rs 3,873; Beta ≈ Rs 4,780.** So an investor in Beta must be prepared to lose about Rs 900 more in a bad month for a *lower* expected return — a clearly worse deal.

Problem 2 — Value at Risk on its own

A stock has mean monthly return 0.9% and monthly SD 5.5%. You invest Rs 2,00,000. Find the worst-case monthly loss at 95% and at 99% confidence.

Solution. VaR $= -\text{mean} + z \cdot \text{SD}$, with $z = 1.645$ (95%) and $z = 2.33$ (99%).

- 95%: $-0.9\% + 1.645(5.5\%) = 8.15\% \Rightarrow$ **Rs 16,295.**
- 99%: $-0.9\% + 2.33(5.5\%) = 11.92\% \Rightarrow$ **Rs 23,830.**

Interpretation: in 19 months out of 20 you should not lose more than ~Rs 16,300; only about 1 month in 100 would exceed ~Rs 23,800.

B. Bond Pricing and Yields (Q2 / Q4b)

Problem 3 — Price a coupon bond

A 5-year bond pays an 8% annual coupon on a Rs 1,000 face value. The market YTM is 10%. Find its price and state whether it trades at a premium or discount.

Solution. $P = \sum_{t=1}^{5} \frac{80}{1.10^t} + \frac{1000}{1.10^5}$. PV of coupons $= 80 \times 3.7908 = 303.26$; PV of face $= 1000/1.6105 = 620.92$.

$$P = 303.26 + 620.92 = \textbf{Rs 924.18}$$

Because the coupon (8%) is **below** the YTM (10%), the bond sells at a **discount** ($P <$ par). Rule: coupon < YTM → discount; coupon = YTM → par; coupon > YTM → premium.

Problem 4 — Current yield and premium/discount

A Rs 1,000 face, 9% annual coupon bond is quoted at Rs 1,050. Find the current yield and say whether YTM is above or below the coupon.

Solution. Current yield $= \frac{\text{annual coupon}}{\text{price}} = \frac{90}{1050} = \textbf{8.57\%}$. Since the price is **above** par, it is a **premium** bond, so the **YTM is below the coupon rate** (YTM < 9%). (Intuitively, you pay more than Rs 1,000 to get the 9% coupons, so your true yield is diluted below 9%.)

Problem 5 — Zero-coupon bond, both directions

(a) A zero-coupon bond of face Rs 1,000 matures in 6 years; YTM is 9%. Find its price. (b) If instead its price is Rs 600, what is its YTM?

Solution. Zero-coupon: $P = \frac{F}{(1+y)^n}$.

- (a) $P = 1000/1.09^6 = \textbf{Rs 596.27}$.
- (b) $y = \left(\frac{1000}{600}\right)^{1/6} - 1 = \textbf{8.89\%}$.

Problem 18 — Perpetuity value and duration

A perpetuity pays Rs 80 per year forever; the market yield is 8%. Find its price and its duration.

Solution. Price $= C/y = 80/0.08 = \textbf{Rs 1,000}$. Duration $= (1+y)/y = 1.08/0.08 = \textbf{13.5 years}$. (A perpetuity has a *finite* duration even though it never matures — because most of its present value arrives in the nearer years.)

C. Duration and Convexity (Q2)

Problem 6 — Build a duration table

A 4-year bond pays a 6% annual coupon on Rs 1,000 face; YTM is 8%. Find its Macaulay duration and modified duration.

Solution. Cash flows: 60, 60, 60, 1060. Discount at 8% and build the table:

t	CF	$PV = CF/1.08^t$	$w_t = PV/P$	$w_t \times t$
1	60	55.56	0.0595	0.0595
2	60	51.44	0.0551	0.1102
3	60	47.63	0.0510	0.1530
4	1060	779.13	0.8344	3.3376
		P = 933.76	**1.000**	**D = 3.660**

Macaulay duration $=$ **3.66 years**. Modified duration $= D/(1+y) = 3.66/1.08 =$ **3.39 years**. *Interpretation:* a 1% rise in yield would cut the price by about 3.39%.

Problem 7 — Convexity correction

A bond has modified duration 6.5 and convexity 70. Yields rise by 1.5%. Estimate the price change (a) with duration only, (b) with the convexity correction.

Solution. $\%\Delta P \approx -MD\,\Delta y + \frac{1}{2}\,\text{Conv}\,(\Delta y)^2$.

- (a) Duration only: $-6.5 \times 0.015 =$ **−9.75%**.
- (b) With convexity: $-9.75\% + \frac{1}{2}(70)(0.015)^2 = -9.75\% + 0.788\% =$ **−8.96%**.

The convexity term (+0.79%) is positive, so duration *overstated* the fall; the true drop is smaller.

Problem 8 — Direction of change (concept)

A bond has a modified duration of 7. State, with reasons, how MD changes if (a) its coupon were higher, (b) its maturity were shorter.

Solution. (a) **Higher coupon → MD falls.** More cash arrives early, so the balance point of the cash flows moves earlier → shorter duration (coupon and duration move inversely). (b) **Shorter maturity → MD falls.** All cash flows arrive sooner → shorter duration (maturity and duration move together).

D. Immunisation (Q3)

Problem 9 — Fund a liability, then rebalance

A firm owes **Rs 50 million in 5 years**. The market rate is **10%**. It will fund this using **3-year zero-coupon bonds** and a **perpetuity** paying annual coupons. (a) PV of the obligation? (b) How much in each instrument? (c) One year later rates rise to 11% — what new mix immunises the (now 4-year) obligation?

Solution.

(a) $PV = \dfrac{50}{1.10^5} = \dfrac{50}{1.6105} =$ **Rs 31.05 million**.

(b) Match duration to the 5-year target. Zero duration = 3; perpetuity duration $= 1.10/0.10 = 11$. Solve

$$w(3) + (1-w)(11) = 5 \;\Rightarrow\; 11 - 8w = 5 \;\Rightarrow\; w = 0.75.$$

So **75% in zeros, 25% in the perpetuity** (feasible because one duration (3) is below the target and the other (11) is above). In money: zeros $= 0.75 \times 31.05 =$ **Rs 23.28 mn**; perpetuity $= 0.25 \times 31.05 =$ **Rs 7.76 mn**.

(c) Now the target is 4 years and $y = 11\%$, so the perpetuity duration becomes $1.11/0.11 = 10.09$ and the zero (one year older) has duration 2. Rebalance:

$$w(2) + (1-w)(10.09) = 4 \;\Rightarrow\; w = \frac{10.09 - 4}{10.09 - 2} = \mathbf{0.753}.$$

So shift to about **75.3% zeros / 24.7% perpetuity.** (The lesson: immunisation must be **re-checked and rebalanced** as time passes and rates move.)

E. Bond Cash Flows Over Time

Problem 10 — Clean vs dirty price

A 10% coupon bond (Rs 1,000 face) pays **semi-annually** (so Rs 50 per coupon). It is bought **4 months** after the last coupon at a quoted price of Rs 970. What does the buyer actually pay?

Solution. Accrued interest $= \text{coupon} \times \dfrac{\text{months since last coupon}}{\text{months in period}} = 50 \times \dfrac{4}{6} =$ **Rs 33.33**.

$$\text{Dirty price} = \text{Clean} + \text{Accrued} = 970 + 33.33 = \textbf{Rs 1,003.33}.$$

The quoted (clean) price is Rs 970, but the **cash the buyer pays is Rs 1,003.33** — the dirty price.

Problem 11 — Horizon (realised) return

You buy a 10-year, 9% annual-coupon bond at par (Rs 1,000) — so YTM = 9%. You will hold it **3 years**, reinvesting coupons at 7%. In 3 years the (now 7-year) bond is forecast to sell at a YTM of 8%. Find the **annualised** return over the 3 years.

Solution.

Step 1 — future value of reinvested coupons (Rs 90 each, reinvested at 7%):

$$90(1.07)^2 + 90(1.07) + 90 = 103.04 + 96.30 + 90 = \text{Rs } 289.34.$$

Step 2 — sale price of a 7-year, 9% bond at YTM 8%: $P = \sum_{t=1}^{7} \dfrac{90}{1.08^t} + \dfrac{1000}{1.08^7} = \text{Rs } 1{,}052.06$.

Step 3 — total and annualise:

$$\text{Total} = 289.34 + 1052.06 = 1341.40; \quad \left(\frac{1341.40}{1000}\right)^{1/3} - 1 = \textbf{10.29\% per year}.$$

Note this exceeds the 9% YTM because the bond is sold at a gain (yields fell to 8%), partly offset by reinvesting coupons at a low 7%.

F. Term Structure of Interest Rates

Problem 12 — Forward rates from the spot curve

Spot (zero) rates are: 1-year 6.0%, 2-year 6.5%, 3-year 7.0%. Find (a) the 1-year forward rate one year from now, and (b) the 1-year forward rate two years from now.

Solution. Use $(1+y_n)^n = (1+y_{n-1})^{n-1}(1+f)$.

- (a) $(1.065)^2 = (1.06)(1+f) \Rightarrow f = \dfrac{1.1342}{1.06} - 1 = \mathbf{7.00\%}$.
- (b) $(1.07)^3 = (1.065)^2(1+f) \Rightarrow f = \dfrac{1.2250}{1.1342} - 1 = \mathbf{8.01\%}$.

The rising forward rates are consistent with an upward-sloping curve (expectations of rising short rates and/or a liquidity premium).

Problem 13 — Expectations vs Liquidity Preference

The expected 1-year rates for the next three years are 5%, 6%, 7%. Liquidity premiums for 1-, 2- and 3-year bonds are 0%, 0.3%, 0.6%. Find the 3-year rate under (a) Pure Expectations, (b) Liquidity Preference.

Solution. The long rate is the **geometric average** of the (premium-adjusted) short rates.

- (a) Pure Expectations: $\left[(1.05)(1.06)(1.07)\right]^{1/3} - 1 = \mathbf{6.00\%}$.
- (b) Liquidity Preference: add each year's premium first: $\left[(1.05)(1.063)(1.076)\right]^{1/3} - 1 = \mathbf{6.30\%}$.

The liquidity premium lifts the long rate by ~0.30%, which is why the curve slopes up even when the expected-rate average alone would be flat.

G. Tests of Market Efficiency (Q4)

Problem 14 — Runs test (weak form)

Daily price changes over 14 days had these signs:

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Test whether the movements are random.

Solution. Count: up-days $n_1 = 8$, down-days $n_2 = 6$. A **run** is an unbroken streak of one sign; counting the streaks gives **actual runs** $R = 8$.

Expected runs under randomness:

$$E(R) = \frac{2n_1n_2}{n_1 + n_2} + 1 = \frac{2(8)(6)}{14} + 1 = 7.86.$$

$$\text{Standard deviation} = \sqrt{\frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2(n_1 + n_2 - 1)}} = 1.76.$$

$$Z = \frac{R - E(R)}{\text{SD}} = \frac{8 - 7.86}{1.76} = \mathbf{0.08}.$$

Z is far inside ± 1.96, so we **cannot reject randomness** — the series is consistent with a **random walk / weak-form efficiency** (no trend to exploit). *(Too few runs would signal trending/momentum; too many, excessive reversals.)*

Problem 15 — Event study (semi-strong form)

A company announces results on Day 2. Its stock's daily returns and the market's returns for five days are below; the market model gives $\alpha = 0.02\%$, $\beta = 1.2$. Compute the abnormal returns and the cumulative abnormal return (CAR).

Day	Stock return	Market return
1	1.2%	0.6%
2 (event)	2.8%	1.0%
3	0.5%	0.4%
4	−0.4%	−0.2%
5	0.9%	0.5%

Solution. Expected return $= \alpha + \beta R_m$; Abnormal return $AR = R_{stock} - (\alpha + \beta R_m)$.

Day	Expected $= 0.02\% + 1.2R_m$	AR	Cumulative CAR
1	0.74%	+0.46%	0.46%
2	1.22%	**+1.58%**	2.04%
3	0.50%	0.00%	2.04%
4	−0.22%	−0.18%	1.86%

Day	Expected $= 0.02\% + 1.2R_m$	AR	Cumulative CAR
5	0.62%	+0.28%	2.14%

Reading it: the big abnormal return (+1.58%) lands on the **event day**, and the CAR then stays roughly flat (2.04% → 2.14%). The price jumped **on the news and then stopped moving** — consistent with **semi-strong efficiency**: the market absorbed the announcement immediately, so you could not profit by trading *after* it. (A CAR that kept drifting up after Day 2 would signal inefficiency; a jump *before* Day 2 would signal leakage.)

H. Market Microstructure

Problem 16 — Bid-ask spread and impact cost

The order book for a share is:

Bids (buy)	Qty	Asks (sell)	Qty
49.50	100	49.80	120
49.40	200	49.90	180
49.20	150	50.10	200

(a) Find the bid-ask spread and mid-price. (b) A market buy order for **250 shares** arrives — find its average execution price and the impact cost.

Solution.

(a) Best bid = 49.50 (highest buy); best ask = 49.80 (lowest sell). Spread $= 49.80 - 49.50 =$ **0.30**. Mid-price $= (49.50 + 49.80)/2 =$ **49.65**.

(b) A 250-share buy fills against the asks: 120 at 49.80, then 130 at 49.90.

$$\text{VWAP} = \frac{120(49.80) + 130(49.90)}{250} = 49.852.$$

Impact cost $= \dfrac{\text{VWAP} - \text{mid}}{\text{mid}} = \dfrac{49.852 - 49.65}{49.65} =$ **0.41%**. This is the hidden cost of demanding immediate liquidity for a large order — it "walks up the book" beyond the best price.

Problem 17 — Fisher real return

A bond returns 11% nominal in a year when inflation is 5%. What is the real return?

Solution. $R_{real} = \dfrac{1+R}{1+i} - 1 = \dfrac{1.11}{1.05} - 1 =$ **5.71%** (the quick approximation $R - i = 6\%$ slightly overstates it).

I. Concept Questions — model answers

These need words, not numbers. Keep each to 4–6 tight lines with one illustration.

Problem 19 — Are markets efficient? Can you beat them?

Markets are **informationally efficient** to a degree: prices quickly reflect available information, so *consistently* earning excess returns after costs is very hard. Efficiency comes in three forms — **weak** (prices reflect past prices, so technical analysis fails), **semi-strong** (all public info, so fundamental analysis fails), **strong** (even private info). Evidence: developed markets are largely weak- and semi-strong-efficient but **not strong-form** (insiders do profit). So for most investors, low-cost **indexing** is rational; but **pockets of inefficiency** persist (Grossman-Stiglitz: someone must be paid to keep prices efficient), leaving room for skilled, informed active managers at the margin.

Problem 20 — Passive or active?

Passive (hold the index) gives the market return with tiny costs and full diversification; **active** (stock-picking) tries to beat it but incurs research and trading costs. If markets are largely efficient, the average active manager **underperforms after costs**, so passive is the sensible default for most investors. Active is justified only where one has a genuine informational or analytical edge (e.g. inefficient small-cap or emerging-market segments). *Decision rule in this course:* compare the active choice's **return-per-unit-risk** to the index's — pick active only if it is clearly higher **and** you can repeat the selection skill.

Problem 21 — Interpreting an inverted yield curve

An **inverted** curve (long yields below short yields) means the market **expects short rates to fall**, which usually happens when investors anticipate an economic **slowdown or recession** and expect the central bank to cut rates. Historically inversion has led recessions by ~4–5 quarters. Mechanism: expecting a downturn, investors buy long bonds now (pushing long yields down) and avoid short ones — flattening then inverting the curve. It is a warning signal, though not infallible, and the liquidity premium normally biases the curve upward, so an inversion is a strong statement.

Problem 22 — What is the maximum you can lose short-selling?

Unlimited. In a short sale you borrow and sell a share hoping to buy it back cheaper; your profit is capped (the price can fall at most to zero, a 100% gain) but your loss is **not** capped, because the price can rise without limit — every rupee it rises is a rupee of loss, and you must eventually buy it back. This asymmetry (limited gain, unlimited loss), plus margin calls and short-squeeze risk, makes shorting far riskier than a normal long position, where the most you can lose is your investment.

*That is one worked problem of every type on your syllabus. If you can reproduce the **method** in each — not memorise these particular numbers — you can handle whatever numbers the exam uses. Show every step, name your criterion, and you will earn the marks. Good luck.*